

Fan Yang

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EDUCATION

- **University of Toronto** Toronto, Canada
Doctor of Philosophy - Electrical and Computer Engineering *September 2024 - Present*
Supervisor: Prof. Ben Liang
- **Tsinghua University** Beijing, China
Master of Engineering - Electronics and Communication Engineering *September 2019 - July 2022*
Supervisor: Prof. Yuhua Dong
- **Sun Yat-sen University** Guangzhou, China
Bachelor of Engineering - Telecommunication Engineering *September 2015 - July 2019*
Supervisor: Prof. Xiang Chen

RESEARCH INTERESTS

Wireless communications, digital signal processing, machine learning.

PUBLICATIONS

1. **F. Yang**, J. Wang, and Y. Dong, "Physical-Layer Security for Indoor VLC Wiretap Systems Under Multipath Reflections," *IEEE Transactions on Wireless Communications*, vol. 21, no. 12, pp. 11179-11192, Dec. 2022.
2. **F. Yang** and Y. Dong, "Joint Probabilistic Shaping and Beamforming Scheme for MISO VLC Systems," *IEEE Wireless Communications Letters*, vol. 11, no. 3, pp. 508-512, Mar. 2022.
3. **F. Yang**, K. Zhang, Y. Zhai, J. Quan, and Y. Dong, "Artificial Noise Design in Time Domain for Indoor SISO DCO-OFDM VLC Wiretap Systems," *Journal of Lightwave Technology*, vol. 39, no. 20, pp. 6450-6458, Oct. 2021.
4. **F. Yang**, K. Zhang, Y. Zhai, and Y. Dong, "Nonlinear HPA Impact on Artificial Noise Aided MISO Secure Systems," in *Proc. IEEE International Conference on Communications (ICC)*, Montreal, QC, Canada, Jun. 2021, pp. 1-6.

HONORS AND AWARDS

- **Outstanding Thesis Award:** Tsinghua University (TOP 5%) *June 2022*
- **National Scholarship:** Tsinghua University (TOP 1%) *October 2021*
- **First Prize:** Sun Yat-sen Undergraduate Electronic Design Contest *December 2015*

TEACHING ASSISTANT EXPERIENCE

- **Probabilistic Reasoning** *Winter 2026*
- **Signals and Systems** *Winter 2026, Winter 2025*
- **Engineering Probability and Statistics** *Fall 2023*
- **Wireless Communication Engineering** *Fall 2021, Fall 2020*

WORK EXPERIENCE

- **The University of British Columbia**
Research Assistant *January 2023 - December 2023*

SKILLS

- **Knowledge:** Communication Theory, Information Theory, Optimization Theory, Signal Processing, Deep Learning.
- **Programming:** Python, C++, MATLAB, GNU Radio